

(*) All the rings will be assumed to be commutative rings with identity if there is no special statement.

Colon Ideal

Definition 0.1. :Let I be an ideal of a ring A and $S \subseteq A$ is a subset of A . Define $I : S := \{x \in A | xs \in I (\forall s \in S)\} = \{x \in A | xS \subseteq I\}$.

Remark 0.1. :If $x, y \in I : S$, then we can check that $x + y \in I : S$, $xy \in I : S$ and $ax \in I : S$ for $\forall a \in A$, which means that $I : S$ is an ideal of A . It is called **colon ideal**.

Example 0.1. :If the subset S satisfies $S \subseteq I$, then $I : S = A$.

Example 0.2. :Let $A = \mathbb{F}[x]$, $I = (x^2)$, $S = \{x\}$, then $I : S = (x)$.

Observation:

(1) Let I be an ideal of the ring A , S_1 and S_2 are two subsets of A such that $S_1 \subseteq S_2$, then we have $I : S_1 \supseteq I : S_2$.

Proof: for $\forall a \in I : S_2$, then $ay \in I (\forall y \in S_2)$, particularly, we have $ay \in I (\forall y \in S_1)$. $\square$

(2) Let I be an ideal of a ring A and $S \subseteq A$ is a subset of A . Let $J = \langle S \rangle$ be the ideal generated by the subset S . Then we have $I : S = I : J$.

Proof: recall that for a commutative ring A with identity and a subset $M \subseteq A$, the ideal generated by M is $\langle M \rangle = \{\sum_i a_i m_i (\text{finite sum}) | a_i \in A, m_i \in M\}$. By (1), we know that $I : S \supseteq I : J$ since $J = \langle S \rangle \supseteq S$. Take $\forall x \in I : S$ and $\forall y \triangleq \sum_i a_i s_i \in J = \langle S \rangle$ ($a_i \in A, s_i \in S$), then $xy = \sum_i xa_i s_i = \sum_i a_i (xs_i) \in I$ since $x \in I : S$ and $s_i \in S$, which means that $x \in I : J$. By the arbitrariness of $x \in I : S$, we get $I : S = I : J$, i.e., $I : S = I : \langle S \rangle$. $\square$

Annihilator

Definition 0.2. :For a ring A and $x \in A$, $(0) : \{x\}$ is called the **annihilator of x** .

Proposition 0.1. :Let I be an ideal of a ring A and $S \subseteq A$ is a subset of A . Then

- (1) $I \subseteq I : S$;
- (2) For two ideals J_1 and J_2 of A , $(I : J_1) : J_2 = I : J_1 J_2$;
- (3) $I(I : S) \subseteq I$;
- (4) For two ideals I_1 and I_2 , $(I_1 \cap I_2) : S = (I_1 : S) \cap (I_2 : S)$.

Proof: easy to check. $\square$

Radical of An Ideal

Definition 0.3. :Let I be an ideal of the ring A , define

$$r(I) := \{x \in A \mid x^n \in I \text{ for some } n \in \mathbb{N}\},$$

called the **radical of I** .

Consider the natural homomorphism

$$\varphi : A \rightarrow A/I$$

$$a \mapsto a + I$$

then by the definition of radical, we can know that $r(I) = \varphi^{-1}(\mathfrak{n}_{A/I})$, where $\mathfrak{n}_{A/I}$ means the nilradical of the quotient ring A/I . Since φ is a ring homomorphism and $\mathfrak{n}_{A/I}$ is an ideal of A/I , thus $r(I) = \varphi^{-1}(\mathfrak{n}_{A/I})$ is an ideal of A .

We have known that

$$\mathfrak{n}_{A/I} = \bigcap_{\alpha} P'_{\alpha},$$

where $\{P'_{\alpha}\}$ is the collection of all prime ideals of A/I .

By set theory we know that for a ring homomorphism $f : A \rightarrow B$ and a collection of ideals $\{I_{\alpha}\}$ of B , $f^{-1}(\cap_{\alpha} I_{\alpha}) = \cap_{\alpha} f^{-1}(I_{\alpha})$.

Thus $\varphi^{-1}(\mathfrak{n}_{A/I}) = \varphi^{-1}(\cap_{\alpha} P'_{\alpha}) = \cap_{\alpha} \varphi^{-1}(P'_{\alpha})$. Since $\varphi^{-1}(P'_{\alpha})$ is a prime ideal of A containing I , we get

$$\varphi^{-1}(\mathfrak{n}_{A/I}) = \bigcap_{\lambda} P_{\lambda}$$

, where $\{P_{\lambda}\}$ is the collection of all prime ideals of A containing I .

Hence we have the proposition

Proposition 0.2. :For an ideal I of a ring A , we have

$$r(I) = \bigcap_{\lambda} P_{\lambda},$$

where $\{P_{\lambda}\}$ is the collection of all prime ideals of A containing I .

Here are some properties about radicals of ideals.

Properties of Radicals

Proposition 0.3. : (1) $I \subseteq r(I)$;

(2) For two ideals I and J , if $I \subseteq J$, then $r(I) \subseteq r(J)$;

(3) $r(r(I)) = r(I)$;

(4) If A is the whole ring, then $r(I) = A \Leftrightarrow I = A$;

(5) For two ideals I, J of the ring A , we have $r(IJ) = r(I \cap J) = r(I) \cap r(J)$;

(6) For two ideals I, J of the ring A , $r(I + J) = r(r(I) + r(J))$;

(7) Let $\mathfrak{p}$ be a prime ideal, then $r(\mathfrak{p}^n) = \mathfrak{p} \ (\forall n \in \mathbb{N})$;

(8) For a maximal ideal $\mathfrak{m}$, $r(\mathfrak{m}) = \mathfrak{m}$.

Proof: (1) trivial;

(2) trivial;

(3) for $\forall x \in r(r(I))$, $\exists m \in \mathbb{N}$ such that $x^m \in r(I)$, hence again $\exists n \in \mathbb{N}$ such that $(x^m)^n = x^{mn} \in I$, which means that $x \in r(I)$ and $r(r(I)) \subseteq r(I)$. By (1), we have $r(I) \subseteq r(r(I))$, thus $r(I) = r(r(I))$;

(4) since A itself is an ideal of A , we have $A \subseteq r(A)$ by (1), therefore $r(A) \subseteq A \subseteq r(A)$ (since $r(A)$ is still an ideal of A), which means $A = r(A)$.

($\Leftarrow$): since $I = A$, we get $r(I) = r(A) = A$;

($\Rightarrow$): since $r(I) = A$, so $1 \in r(I)$, which implies that $\exists n \in \mathbb{N}$ such that $1 = 1^n \in I$, thus $I = A$;

(5) it's clear that $r(I \cap J) \subseteq r(I) \cap r(J)$; for $\forall x \in r(I) \cap r(J)$, $\exists n, m \in \mathbb{N}$ such that $x^n \in I$ and $x^m \in J$, then $x^{n+m} \in I \cap J$, hence $x \in r(I \cap J)$ and $r(I) \cap r(J) \subseteq r(I \cap J)$. Therefore, $r(I \cap J) = r(I) \cap r(J)$;

Since $IJ \subseteq I \cap J$, we get $r(IJ) \subseteq r(I \cap J)$. For $\forall x \in r(I \cap J)$, then $\exists n \in \mathbb{N}$ such that $x^n \in I \cap J$, i.e., $x^n \in I$ and $x^n \in J$, thus $x^{2n} = x^n \cdot x^n \in IJ$, hence $x \in r(IJ)$ and $r(I \cap J) \subseteq r(IJ)$, which means $r(IJ) = r(I \cap J) = r(I) \cap r(J)$;

(6) it's clear that $r(I) + r(J) \subseteq r(I + J)$, then we have $r(r(I + J)) = r(I + J) \supseteq r(r(I) + r(J))$. Since $I \subseteq r(I)$ and $J \subseteq r(J)$, $I + J \subseteq r(I) + r(J)$, then $r(I + J) \subseteq r(r(I) + r(J))$. Thus $r(I + J) = r(r(I) + r(J))$;

(7) for $\forall x \in \mathfrak{p}$, $x^n \in \mathfrak{p}^n \subseteq r(\mathfrak{p}^n)$, hence $\mathfrak{p} \subseteq r(\mathfrak{p}^n)$. We have known that $r(\mathfrak{p}^n)$ equals to the intersection of all prime ideals containing $\mathfrak{p}^n$, since $\mathfrak{p} \supseteq \mathfrak{p}^n$, therefore we get $r(\mathfrak{p}^n) \subseteq \mathfrak{p}$;

(8) by (1), we have $\mathfrak{m} \subseteq r(\mathfrak{m})$. By (4), we have $r(\mathfrak{m}) = A \Leftrightarrow \mathfrak{m} = A$, since $\mathfrak{m}$ is a maximal ideal, then $\mathfrak{m} \neq A$ and $r(\mathfrak{m}) \neq A$, hence $\mathfrak{m} = r(\mathfrak{m})$. $\square$

Extension and Contraction of Ideals

Definition 0.4. :Let $f : A \rightarrow B$ be a ring homomorphism, I is an ideal of A and J is an ideal of B . Define

$$I^e := \left\{ \sum_i b_i f(x_i) \text{ (finite sum)} \mid b_i \in B, x_i \in I \right\},$$

i.e., I^e is the ideal of B generated by $f(I)$, the image of I . I^e is called an **extension** of I . Define

$$J^c := f^{-1}(J),$$

clearly, J^c is an ideal of A , it is called a **contraction** of J .

Remark 0.2. :By the definition above, we can give a correspondence

$$I \rightsquigarrow f(I)B =: I^e$$

$$J \rightsquigarrow f^{-1}(J) =: J^c$$

Example 0.3. :Consider the inclusion homomorphism

$$\mathbb{Z} \hookrightarrow \mathbb{Q},$$

then for the ideal $n\mathbb{Z}$ of $\mathbb{Z}$ and the ideal $\mathbb{Q}$ itself, we have $(n\mathbb{Z})^e = \mathbb{Q}$ and $\mathbb{Q}^c = \mathbb{Z}$

Example 0.4. :Consider the inclusion homomorphism

$$\mathbb{Z} \hookrightarrow \mathbb{Z}[x],$$

then for the ideal $n\mathbb{Z}$ of $\mathbb{Z}$, we have $(n\mathbb{Z})^e = n\mathbb{Z}[x]$ and $(n\mathbb{Z}[x])^c = n\mathbb{Z}[x] \cap \mathbb{Z} = n\mathbb{Z}$.

Example 0.5. :Suppose A is a subring of B , consider the inclusion homomorphism

$$A \hookrightarrow B,$$

then for any ideal I of A and any ideal J of B , we have $I^e = IB$ and $J^c = J \cap A$.

Properties of Extension and Contraction

Proposition 0.4. :Let $f : A \rightarrow B$ be a ring homomorphism, I is an ideal of A and B is an ideal of B , then we have

$$(1) (I^e)^c \supseteq I;$$

$$(2) (J^c)^e \subseteq J;$$

$$(3) I^{ece} = I^e, \text{ here } I^{ece} \text{ is short for } ((I^e)^c)^e;$$

$$(4) J^{cec} = J^c;$$

(5) If C is the set of all contracted ideals of A and E is the set of all extended ideals of B , then

$$C = \{I \mid I \text{ is an ideal of } A, I^{ec} = I\}, E = \{J \mid J \text{ is an ideal of } B, J^{ce} = J\}.$$

Proof:(1)For $\forall x \in I$, $f(x) \in f(I) \subseteq \langle f(I) \rangle = I^e$, hence $x \in f^{-1}(I^e) = (I^e)^c$, therefore $I \subseteq (I^e)^c$;

(2)By definition, $(J^c)^e = \{\sum_i a_i s_i | s_i \in f(J^c), a_i \in B\}$. For $\forall y \in J^c = f^{-1}(J)$, we have $f(y) \in J$, hence $f(J^c) \subseteq J$, which means that for every $s_i \in f(J^c)$, $s_i \in J$. Thus for every element $x \triangleq \sum_i a_i s_i$, we have $x \in J$, therefore $(J^c)^e \subseteq J$;

(3)By (1), we have $I^{ec} \supseteq I$, hence $I^{ece} = (I^{ec})^e \supseteq I^e$. By (2), we have $I^{ece} = (I^e)^{ce} \subseteq I^e$. Thus $I^{ece} = I^e$;

(4)similar to (3);

(5)If $I \in C$, then $\exists J$ such that $I = J^c$, i.e., $I = J^c = J^{cec} = (J^c)^{ec} = I^{ec}$. Conversely, if $I = I^{ec}$, then $I = (I^e)^c$, so $I \in C$; For E , the proof is similar. $\square$